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Integrating predictive coding with reservoir computing in spiking neural network model of cultured neurons

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Abstract: The field of physical reservoir computing, which harnesses the dynamics of physical systems for information processing, is undergoing continuous development. The use of cultured neuronal networks as a physical system within physical reservoir computing has played a crucial role in clarifying the connection between neuronal network structure and information processing. In this study, we introduce a model that combines predictive coding and reservoir computing within cultured neurons, serving as a model for sensory information processing. Our findings indicate that the interplay between neuronal network structure and the number of neurons has a significant influence on sensory information processing. This research lays the groundwork for future inquiries into information processing within biological neuronal networks.

Key Words: reservoir computing, predictive coding, cultured neuron, modular organization, computational neuroscience

1. Introduction

Sensory information processing and perception are not merely bottom-up processes of sensory signals. The brain's internal states exert a significant influence on the perception process through top-down processing. The internal network generates attentional modulation for sensory processing and makes predictions about incoming sensory signals. The robust and flexible processing of perception is achieved through the synergy between bottom-up and top-down processing. However, the mechanisms underlying the dynamics of perception remain largely unexplored.

As a computational model of the dynamical process of perception, Katori previously proposed a neural network model based on the predictive coding and reservoir computing [1]. Katori's model uses reservoir computing as a generative model for predictive coding, which has been proposed as one of the mechanisms of perception [2]. The internal network dynamics of Katori's model are such that the network reproduces the specific multidimensional time series course of the sensory signal. In addition, Katori's model is trained so that prediction errors are sent to higher-order networks to trigger the internal network dynamics.

Reservoir computing was initially introduced as a computational model that employed artificial neural networks or spiking neural networks as the reservoir layer. Over time, the field of physical reservoir computing has emerged, exploiting the intricate high-dimensional dynamics of various physical systems as the "reservoir" to perform tasks such as pattern recognition, pattern generation, and time-series prediction [3]. One of the physical systems that have been investigated is biological neurons cultured in vitro [4]. The major advantages of using cultured neuronal networks are threefold: they enable the use of biological neurons that constitute the brain of animals, they provide a high degree of control over network structure, and they simultaneously provide direct access for recording the network activity at a single-cell resolution. These properties, particularly the latter two, are typically challenging to achieve in experiments with the actual brain and render cultured neurons an ideal counterpart to computational models.

Leveraging such benefits, Sumi et al. recently revealed the significant influence of the modular structure in the computational properties of biological neuronal networks [5]. This was achieved by implementing a time-series classification task in physical reservoir of cultured neurons. A modular structure is a topology characterized by multiple densely interconnected sub-populations (referred to as modules) with relatively sparse connections between these modules [6] and is known to be evolutionarily conserved in the nervous systems of animals [7]. The modular structure enabled the neuronal network to display increased complexity in both the spontaneous [8] and noise-driven states [9] compared to a network composed of a single module. This modularity also provided functional advantages for physical reservoir computing in cultured neurons by increasing complexity of the network response to diverse inputs [5]. However, how modular topology influences the performance of the neuronal reservoir in the context of predictive coding, a more biologically plausible model, remains elusive.

In this study, we present a model that integrates a spiking neural network acting as a reservoir that functions as a network for predictive coding. We have previously realized a framework for pattern recognition that converts spatiotemporal patterns of input stimuli into corresponding labels through a network of cultured neurons [5]. The target of the present research is to emulate the mechanisms of sensory information processing that occur in the cerebral cortex. Specifically, this is based on a predictive coding mechanism that uses the brain's dynamic network to predict and reconstruct incoming sensory stimuli, with prediction errors updating the internal state. Sensory stimuli, in the form of spatiotemporal patterns, are reconstructed through a neuronal network functioning as a reservoir. Implementing these systems poses challenges due to the immediate feedback requirement from the reservoir. The primary aim of our model is to simulate the function of biological neuronal networks in culture and to establish a basic framework for constructing models of sensory information processing in cultured systems. We evaluated the performance of this model based on the fundamental task of predicting periodic functions. Moreover, during the predictive phase, the network's output is fed back into its network, allowing it to operate as an autonomous system. Using this feedback mechanism, we perform predictions of time-series patterns. This study contributes to the elucidation of information processing mechanisms in biological neuronal networks and to the development of bio-inspired artificial intelligence.

2. Method

Our model consists of an input layer, a prediction error layer, a dynamical reservoir, and a prediction layer (Fig. 1).

The dynamical reservoir is constructed using leaky integrate-and-fire neurons, supplemented with

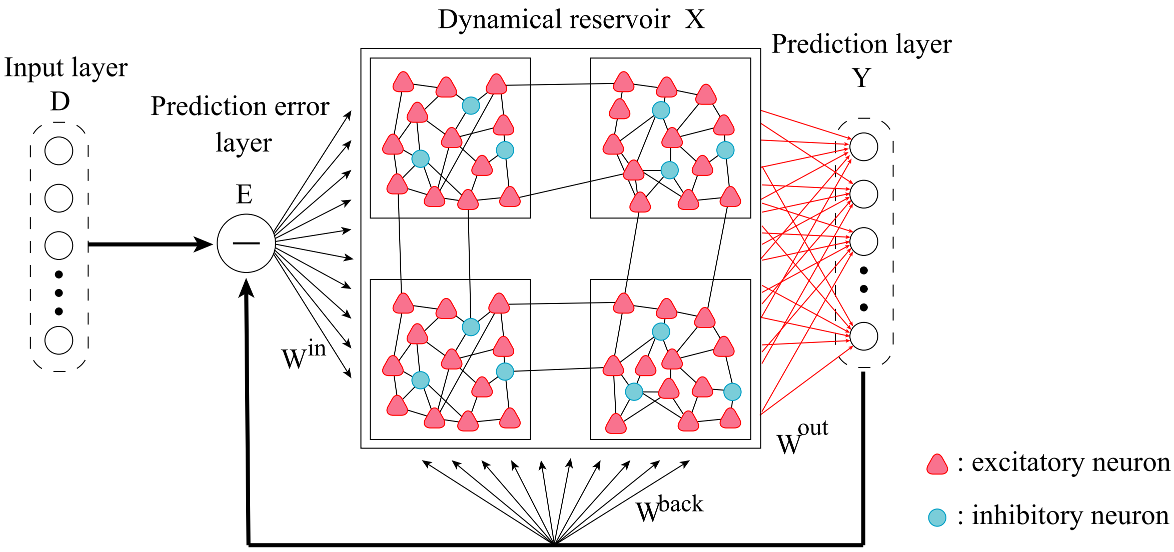

Fig. 1. Architecture of the model with predictive coding and reservoir computing in cultured neurons.

a K^+ current that is dependent on the intracellular Ca^{2+} concentration and a refractory current, to mimic the behavior of cultured neurons [8, 10, 11]. By incorporating of the K^+ current, the model evaluates Ca^{2+} at each time step, allowing us to draw comparisons with fluorescence calcium imaging used in biological experiments [5]. The ratio of excitatory to inhibitory neurons is set to $4:1$, reflecting the values observed in the cerebral cortex [12]. The neurons in the dynamical reservoir are grouped into modules, or subpopulations of neurons, arranged in a $M \times M$ lattice. Neurons within a given module are connected with a probability p_{intra}, whereas those belonging to different modules are connect with a probability p_{inter}. The membrane potential dynamics in excitatory and inhibitory neurons are defined by the following equations:

$$\tau^{\mathrm{E}}\frac{dV^{\mathrm{E}}(t)}{dt} = E_{\mathrm{L}}^{\mathrm{E}} - V^{\mathrm{E}}(t) + R_{\mathrm{in}}^{\mathrm{E}}(I^{\mathrm{E}}(t) + I^{\mathrm{I}}(t) + I_{\mathrm{K(Ca)}}(t) + I_{\mathrm{ref}}(t) + I_{\mathrm{noise}}(t) + I_{\mathrm{back}}(t)), \tag{1}$$

$$\tau^{\mathrm{I}}\frac{dV^{\mathrm{I}}(t)}{dt} = E_{\mathrm{L}}^{\mathrm{I}} - V^{\mathrm{I}}(t) + R_{\mathrm{in}}^{\mathrm{I}}(I^{\mathrm{E}}(t) + I^{\mathrm{I}}(t) + I_{\mathrm{noise}}(t) + I_{\mathrm{back}}(t)), \tag{2}$$

where the E and I superscripts represent excitatory and inhibitory neurons, respectively. τ is the membrane time constant, E_{L} is the resting membrane potential and R_{in} is the input resistance. The excitatory and inhibitory synaptic currents are defined by the following equations:

$$I^{\mathrm{E,I}}(t) = g^{\mathrm{E,I}}(t)(V_{\mathrm{R}}^{\mathrm{E,I}} - V^{\mathrm{E,I}}(t)), \tag{3}$$

where V_{R} is the reversal potential, t_j^a is the a-th spike arrival time of neuron j, and Δ_j is synaptic transmission delay. The excitatory and inhibitory synaptic conductances are defined by the following equations:

$$\frac{dg^{\mathrm{E,I}}(t)}{dt} = -\frac{g^{\mathrm{E,I}}}{\tau^{\mathrm{sE,sI}}} + \bar{g}^{\mathrm{E,I}}\sum_j\sum_a \delta(t - (t_j^a + \Delta_j)), \tag{4}$$

where τ^{s} is the synaptic time constant, and $\bar{g}$ is the maximum conductance. δ is the Dirac delta function. The noise currents are defined by the following the Ornstein-Uhlenbeck (OU) process [13]:

$$\frac{dI_{\mathrm{noise}}(t)}{dt} = \frac{\mu - I_{\mathrm{noise}}(t)}{\tau_{\mathrm{noise}}} + \sigma\xi, \tag{5}$$

where ξ follows a normal distribution with mean 0 and variance 1. μ is the mean, σ is the noise intensity, and τ_{noise} is the time constant of the OU process. The Ca^{2+}-dependent K^+ current is defined by the following equations:

$$I_{\mathrm{K(Ca)}}(t) = g_{\mathrm{K(Ca)}}c(t)(E_{\mathrm{K}} - V^{\mathrm{E}}(t)), \tag{6}$$

$$\frac{dc(t)}{dt} = c_{\text{step}} \sum_a \delta(t - t^a) - \frac{c(t)}{\tau_{\text{Ca}}}, \tag{7}$$

where $g_{\text{K(Ca)}}$ is the potassium current conductance, $c(t)$ is the intracellular Ca^{2+} concentration, E_{K} is the K^+ current reversal potential, c_{step} is the Ca^{2+} unit inflow, τ_{Ca} is the Ca^{2+} channel time constant. The refractory current is defined by the following equations:

$$I_{\text{ref}}(t) = -g_{\text{ref}} \left(1 + \frac{t - t^a}{\tau_{\text{ref}}}\right)^{-1} P_{\gamma a}(t - t^a)(V^{\text{E}}(t) - V_{\text{reset}}), \tag{8}$$

$$P_{\gamma a}(t - t^a) = \begin{cases} 1 & \text{for} \quad t^a < t < t^{a+1}, \\ 0 & \text{otherwise}, \end{cases} \tag{9}$$

where g_{ref} is the refractory current conductance, τ_{ref} is the refractory current time constant, V_{reset} is the reset potential. The feedback current is defined by the following equations:

$$I_{\text{back}}(t) = I_{\text{steady}} + \alpha W^{\text{in}} \boldsymbol{e}(t) + W^{\text{back}} \boldsymbol{y}(t), \tag{10}$$

where α is the error-driven coefficient, I_{steady} is the steady current, $W^{\text{in}} \in \mathbb{R}^{N_{\text{x}} \times N_{\text{d}}}$ was the prediction error connection weight matrix, N_{x} is the number of neurons, N_{d} is the dimension of sensory input, $\boldsymbol{e}(t) \in \mathbb{R}^{N_{\text{d}}}$ is the prediction error, and $W^{\text{back}} \in \mathbb{R}^{N_{\text{x}} \times N_{\text{d}}}$ is the prediction connection weight matrix. The model runs in two modes, i.e., the error-driven and the free-running [14]. In the error-driven mode, the prediction error is fed back to the reservoir and the network is trained to minimize subsequent prediction errors. In the free-running mode, the reservoir operates solely on feedback input from the prediction layer, without incorporating the prediction error. The settings of $\alpha = 1$ and 0 make the model in the error-driven and the free-running modes, respectively. The prediction error $\boldsymbol{e}(t)$ is calculated as $\boldsymbol{d}(t) - \boldsymbol{y}(t)$, where $\boldsymbol{d}(t) \in \mathbb{R}^{N_{\text{d}}}$ is the sensory input, and $\boldsymbol{y}(t) \in \mathbb{R}^{N_{\text{d}}}$ is the prediction. The maximum and minimum values of $\alpha W^{\text{in}} \boldsymbol{e}(t)$ and $W^{\text{back}} \boldsymbol{y}(t)$ are I_{steady} and $-I_{\text{steady}}$, respectively.

The adjacency matrices W^{in} and W^{back} are defined based on P_{input}, β, and γ, where P_{input} is the proportion of neurons receiving inputs to total neurons and β and γ are the coefficients of the prediction error feedback weights and prediction feedback weights, respectively. Neurons that receive inputs are selected randomly.

The dynamical reservoir $\boldsymbol{x}(t) \in \mathbb{R}^{N_{\text{x}}}$ describes low-pass filtered spike trains as defined the following function:

$$\frac{d\boldsymbol{x}_j(t)}{dt} = x_{\text{step}} \sum_a \delta(t - t_j^a) - \frac{\boldsymbol{x}_j(t)}{\tau_{\text{x}}}, \tag{11}$$

where j is the neuron index, x_{step} is the width of the rise of $\boldsymbol{x}(t)$, t_j^a is the a-th spike arrival time of neuron j and τ_{x} is the reservoir time constant.

The prediction $\boldsymbol{y}(t)$ is generated according to the following equation:

$$\boldsymbol{y}(t) = W^{\text{out}} \boldsymbol{x}(t), \tag{12}$$

where $W^{\text{out}} \in \mathbb{R}^{N_{\text{d}} \times N_{\text{x}}}$ is the output connection weight matrix. The output connections from the dynamical reservoir W^{out} is updated with first-order and controlled error learning algorithm as following [14, 15]:

$$\boldsymbol{v}(t) = P(t)\boldsymbol{x}(t+1), \tag{13}$$

$$P(t+1) = P(t) - \frac{\boldsymbol{v}(t)\boldsymbol{v}^T(t)}{1 + \boldsymbol{v}^T(t)\boldsymbol{x}(t)}, \tag{14}$$

$$W^{\text{out}}(t+1) = W^{\text{out}}(t) + \boldsymbol{e}(t)\boldsymbol{x}^T(t)P(t), \tag{15}$$

The initial value of $P(t) \in \mathbb{R}^{N_{\text{x}} \times N_{\text{x}}}$ is

$$P(0) = \frac{\boldsymbol{I}}{\lambda}, \tag{16}$$

where the matrix $\boldsymbol{I}$ is an identity matrix and λ is a scaling parameter.

The numerical experiment conducted in this study focuses on the prediction of a sine wave, which is provided as a sensory input. The sine wave has an amplitude of 800 pA, an offset of 1000 pA, and a period of 4000 ms, unless otherwise noted. The performance was evaluated during both the prediction during the error-driven mode and that during the free-running mode, based on the root mean squared error (RMSE) and the maximum value of a cross-correlation function between the sensory input and prediction. The RMSE is normalized by dividing it by the amplitude of the input sine wave. The total simulation time is 40000 ms, consisting of a learning phase (from 0 to 20000 ms), the error-driven phase (from 20000 to 30000 ms), and the free-running phase (from 30000 to 40000 ms). Table I. summarizes the parameters used in the experiment.

Table I. Parameter.

Parameter	Sign	Value	Unit
Number of sensory input	N_{d}	1	
Number of neurons	N_{x}	$100-1000$	
Proportion of neurons receiving inputs to total neurons	P_{input}	0.1	
Number of modules	M	$1-5$	
Connection probability	p_{intra}	0.25	
	p_{inter}	0.01	
Membrane time constant	τ^{E}	20	ms
	τ^{I}	10	ms
Resting membrane potential	$E_{\mathrm{L}}^{\mathrm{E}}$	-74	mV
	$E_{\mathrm{L}}^{\mathrm{I}}$	-70	mV
Input resistance	$R_{\mathrm{in}}^{\mathrm{E}}$	40	MΩ
	$R_{\mathrm{in}}^{\mathrm{I}}$	50	MΩ
Reversal potential	$V_{\mathrm{R}}^{\mathrm{E}}$	0	mV
	$V_{\mathrm{R}}^{\mathrm{I}}$	-80	mV
Conductance coefficient	$\bar{g}^{\mathrm{E}}$	3	nS
	$\bar{g}^{\mathrm{I}}$	6	nS
Synaptic time constant	τ^{sE}	2	ms
	τ^{sI}	5	ms
Mean of the OU process	μ^{E}	350	pA
	μ^{I}	250	pA
Noise intensity of the OU process	σ^{E}	200	pA
	σ^{I}	100	pA
Time constant of the OU process	τ_{noise}	5.3	ms
Potassium current conductance	$g_{\mathrm{K(Ca)}}$	10	nS/μM
K^+ current reversal potential	E_{K}	-75	mV
Ca^{2+} unit inflow	c_{step}	0.1	μM
Ca^{2+} channel time constant	τ_{Ca}	2700	ms
Refractory current conductance	g_{ref}	150	nS
Refractory current time constant	τ_{ref}	12	ms
Reset potential	V_{reset}	-60	mV
Steady current	I_{steady}	50	pA
Error feedback coefficient	β	0.05	
Prediction feedback coefficient	γ	0.4	
Width of the rise of $\boldsymbol{x}$	x_{step}	0.1	
Reservoir time constant	τ_{x}	200	ms
Scaling parameter	λ	10	
Sensory input current time step	ds	10	ms
Simulation time step	dt	0.2	ms

3. Results

Figures 2, 3, and 4 depict the model response to sine waves with periods of 2000, 4000, and 6000 ms, respectively. The estimated values of average firing rate, RMSE, and cross-correlation with sensory

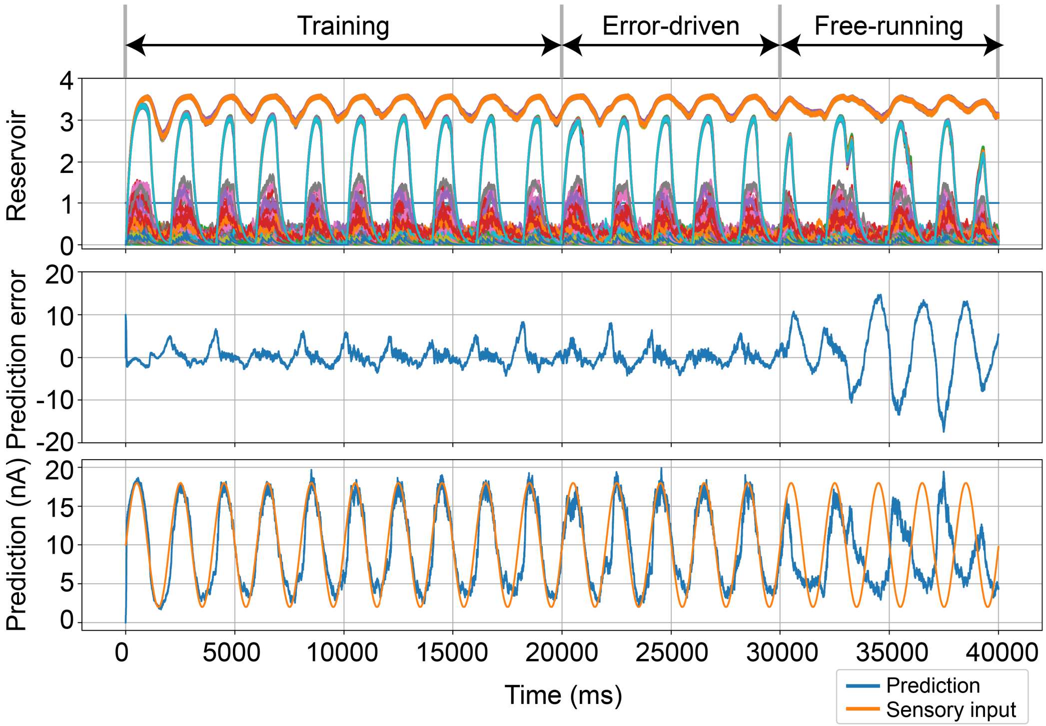

Fig. 2. Response of the model with predictive coding and reservoir computing in cultured neurons to a sine wave. The period of a sine wave is 2000 ms. $N_x = 500$, $M = 2$. The panels show the time series of the state of the dynamical reservoir (upper), the prediction error (middle), and the prediction (lower).

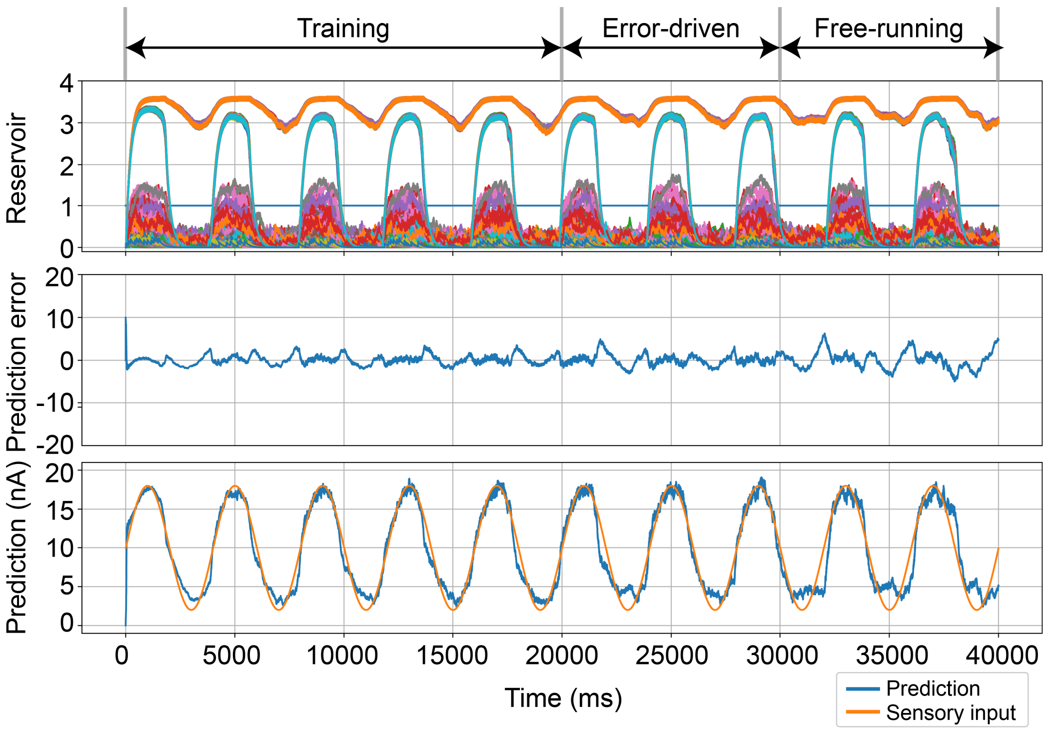

Fig. 3. Response of the model with predictive coding and reservoir computing in cultured neurons to a sine wave. The period of a sine wave is 4000 ms. $N_x = 500$, $M = 2$. The panels show the time series of the state of the dynamical reservoir (upper), the prediction error (middle), and the prediction (lower).

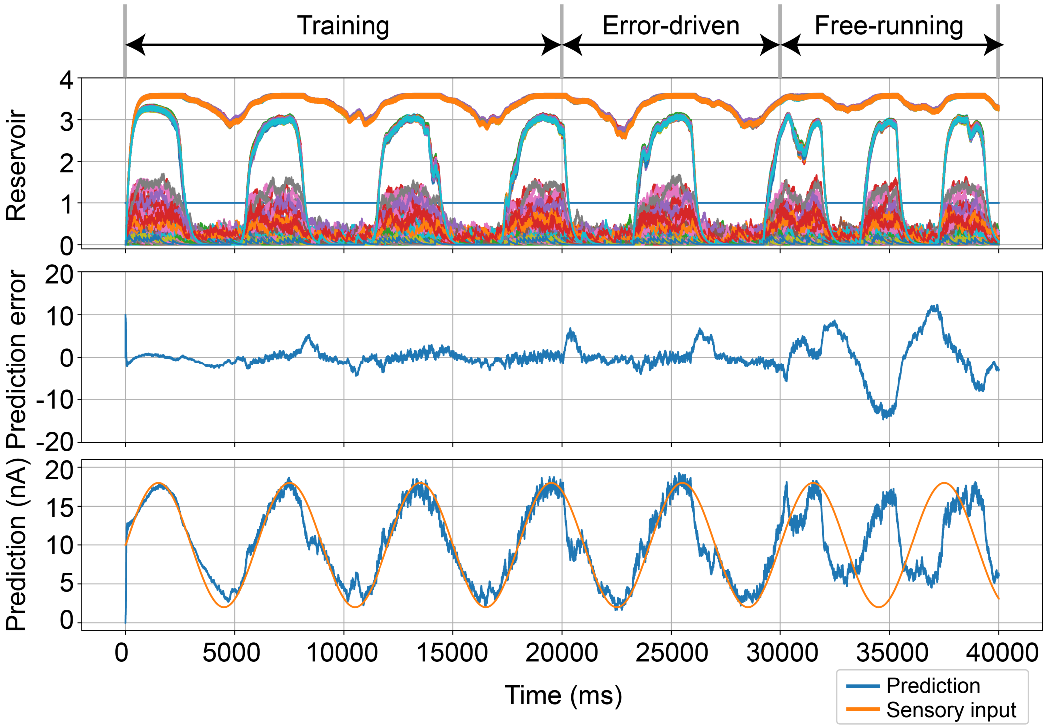

Fig. 4. Response of the model with predictive coding and reservoir computing in cultured neurons to a sine wave. The period of a sine wave is 6000 ms. $N_{\mathrm{x}} = 500$, $M = 2$. The panels show the time series of the state of the dynamical reservoir (upper), the prediction error (middle), and the prediction (lower).

Table II. Average firing rate of Fig. 2.

Average firing rate	Error-driven mode	Free-running mode	Unit
All neurons	10.7	3.4	spk/s
Excitatory neurons	6.2	1.9	spk/s
Inhibitory neurons	28.4	9.3	spk/s

Table III. Average firing rate of Fig. 3.

Average firing rate	Error-driven mode	Free-running mode	Unit
All neurons	11.2	3.4	spk/s
Excitatory neurons	6.8	1.9	spk/s
Inhibitory neurons	28.7	9.3	spk/s

Table IV. Average firing rate of Fig. 4.

Average firing rate	Error-driven mode	Free-running mode	Unit
All neurons	10.9	4.1	spk/s
Excitatory neurons	6.5	2.6	spk/s
Inhibitory neurons	28.6	10.0	spk/s

input are summarized in Tables II, III, IV, and V. In the error-driven mode, the model accurately predicts the sine waves with the same sensory input periods. However, in the free-running mode, the predicted periods to the sin waves with 2000 and 4000 ms periods are 1.5 and 0.7 times as long as their original periods in both modes, the model successfully predicts the 4000 ms input.

Figure 5 shows the dependence of the period of sine wave of RMSE, the maximum value of a cross-correlation function for sensory input and prediction, and spike rate. During the error-driven mode, the RMSE is minimized and the maximum value of the cross-correlation is maximized around the period of 4000 ms. In free-running mode, the RMSE is extremely low and the maximum value of the

Table V. Evaluation metrics.

Evaluation metrics	Period of sine wave	Error-driven mode	Free-running mode
RMSE	2000 (ms)	0.534	7.207
RMSE	4000 (ms)	0.214	0.578
RMSE	6000 (ms)	0.553	5.377
Maximum value of a cross-correlation function	2000 (ms)	0.962	0.475
Maximum value of a cross-correlation function	4000 (ms)	0.973	0.948
Maximum value of a cross-correlation function	6000 (ms)	0.933	0.386

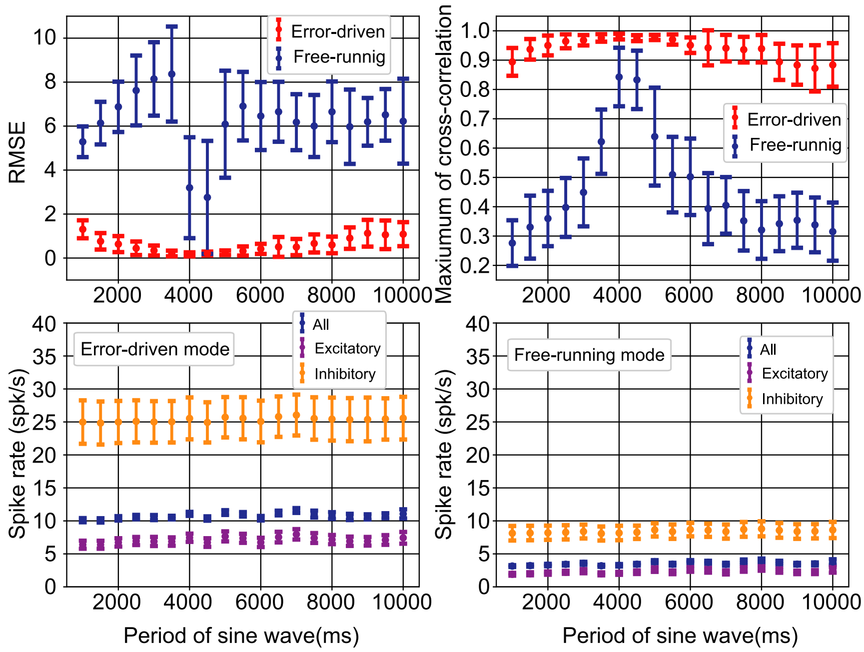

Fig. 5. The changes in RMSE, the maximum value of a cross-correlation function for sensory input and prediction, and the spike rate with respect to the period of sine wave. The mean is represented by circles and the variance by error bars. The evaluation was done with 100 simulations. $N_x = 500$, $M = 2$.

cross-correlation function is extremely high when the period of sine wave is 4000 ms and 4500 ms. The spike rate remained relatively constant irrespective of the period of the input sine wave.

Figure 6 shows the dependence of the reservoir time constant of RMSE, the maximum value of a cross-correlation function for sensory input and prediction, and spike rate. During the error-driven mode, as the value of the reservoir time constant is increased, the maximum value of a cross-correlation function increases. During the free-running mode, an optimal reservoir time constant ranges between 150 and 200 ms, where RMSE is low, and the maximum value of the cross-correlation function is high. The reservoir time constant little affect the spike rate.

Figure 7 shows the variation of RMSE, the maximum value of a cross-correlation function for sensory input and prediction, and spike rate with respect to the proportion of inhibitory neurons. During error-driven mode, the RMSE is lower and the maximum cross-correlation function is larger when the proportion of inhibitory neurons ranges from 0.2 to 0.5. In free-running mode, the closer the proportion of inhibitory neurons is to 0.2, the lower the RMSE and the larger the the maximum value of a cross-correlation function. The larger proportion of inhibitory neurons results in a smaller spike rate of inhibitory and excitatory neurons and a larger overall spike rate.

Figure 8 shows the variation of RMSE and the maximum value of a cross-correlation function for sensory input and prediction with respect to the number of modules and the number of neurons. In the error-driven mode, RMSE is higher and the maximum value of a cross-correlation function is relatively low when the number of modules is 1. The similar trend is observed when the number of neurons is 100. During the free-running mode, lower RMSE values are recorded when the number of modules is 1 and the number of neurons is 100, when the number of modules is 2×2 and the number

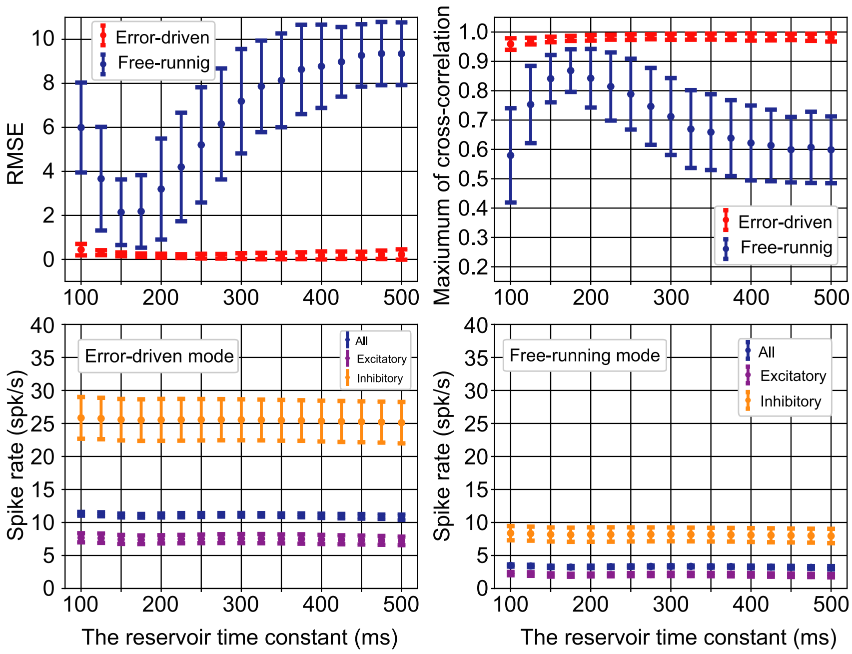

Fig. 6. The changes in RMSE, the maximum value of a cross-correlation function for sensory input and prediction, and the spike rate with respect to the reservoir time constant. The mean is represented by circles and the variance by error bars. The evaluation was done with 100 simulations. The period of a sine wave is 4000 ms. $N_\mathrm{x} = 500$, $M = 2$.

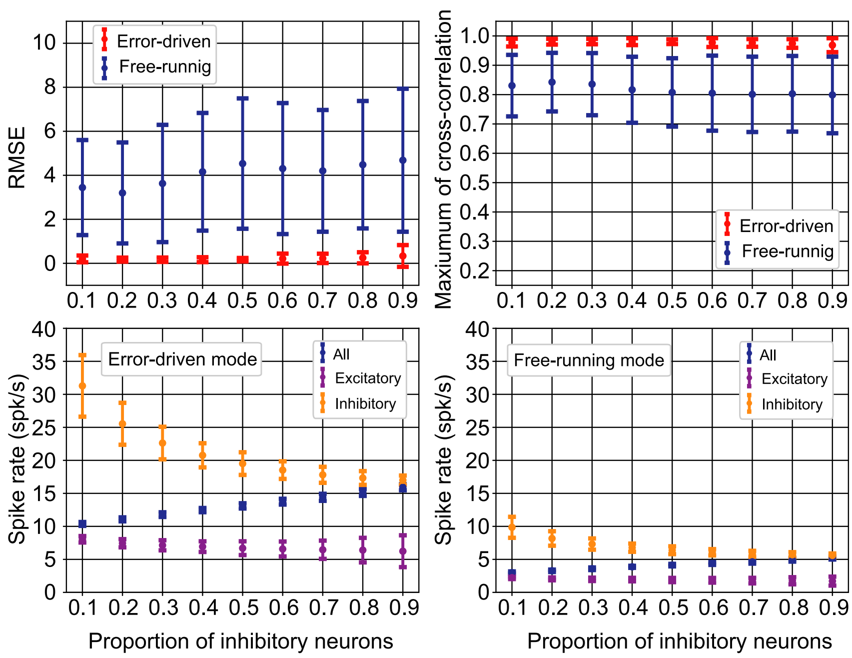

Fig. 7. The changes in RMSE, the maximum value of a cross-correlation function for sensory input and prediction, and the spike rate with respect to the proportion of inhibitory neurons. The mean is represented by circles and the variance by error bars. The evaluation was done with 100 simulations. The period of a sine wave is 4000 ms. $N_\mathrm{x} = 500$, $M = 2$.

of neurons is 500, and when the number of modules is 3×3 and the number of neurons is 1000. Also, when the number of modules is 1 and the number of neurons is 100, the maximum value of a cross-correlation function is higher when the number of modules is 3×3 and the number of neurons is 1000.

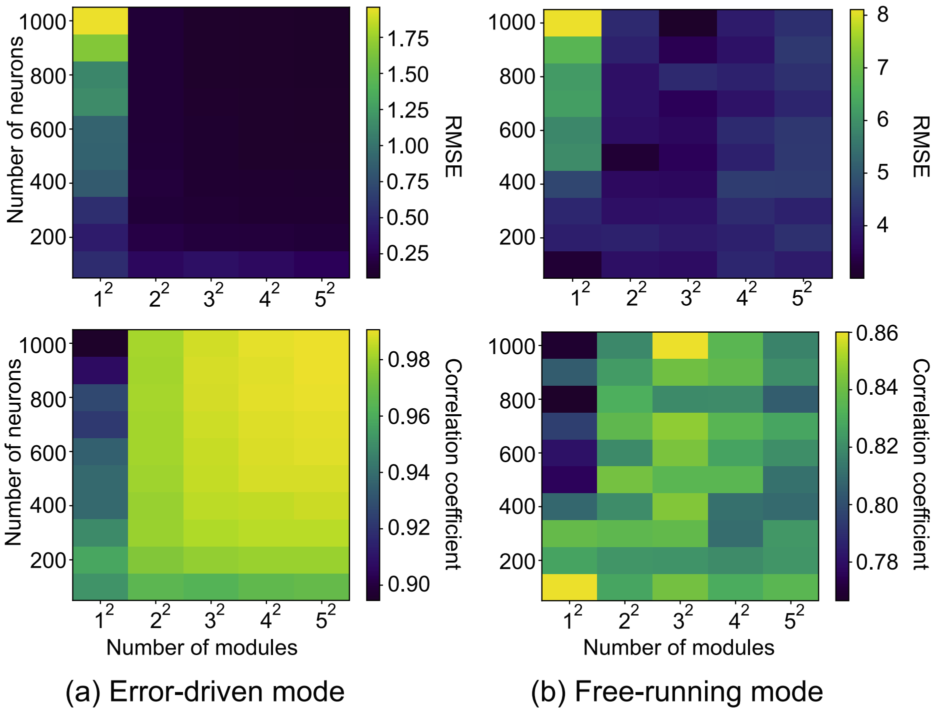

Fig. 8. The changes in RMSE and the maximum value of a cross-correlation function for sensory input and prediction with respect to the number of modules and neurons. The evaluation was done with 100 simulations. The period of a sine wave is 4000 ms.

4. Conclusion

The present study proposed a dynamical model of perception based on the concept of reservoir computing and predictive coding using spiking neuron models. The model exhibited the highest prediction accuracy when the period of the input sine wave was between 4000 ms and 4500 ms. Several factors, including the number of neurons, the number of modules, and the Ca^{2+} channel time constant, may be responsible for the high prediction accuracy in this range. When information is represented through firing activity within a neural network, the duration for which the impact of the firing is maintained becomes crucial. This study revealed that the prediction accuracy of the model improves when the reservoir time constant lies within the range of approximately 150–200 ms. There exists an appropriate range for the number of neurons per module, and the study determined that the optimal number of neurons per module is approximately 120. This study provides a foundation for elucidating the dynamical basis of perception and information processing in the brain, and offers guidelines for designing biological experiments using cultured neuronal networks.

Acknowledgments

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